PQLI
Guide Series

Product Quality Lifecycle Implementation (PQLI®) from Concept to Continual Improvement

Part 3 – Change Management System as a Key Element of a Pharmaceutical Quality System

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Preface

The ISPE Product Quality Lifecycle Implementation (PQLI) Good Practice Guide (GPG): Change Management System is Part 3 of the ISPE Guide Series: PQLI from Concept to Continual Improvement, which is a product of the ISPE PQLI program.

The modern pharmaceutical quality system described in ICH Q10 is a holistic approach which helps to facilitate the consistent development and production of high quality pharmaceutical products. The ISPE PQLI GPG: Change Management System describes potential product lifecycle approaches to the change management system element of a pharmaceutical quality system. These approaches can help to achieve the opportunities described in ICH Q10.

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Table of Contents

1 Introduction

1.1 Background

The modern Pharmaceutical Quality System (PQS) described in ICH Q10 [1] is a holistic approach which helps to facilitate the consistent development and production of high quality pharmaceutical products. It also aims to support innovation and continual improvement of products, processes, and methodologies in accordance with the concepts described in ICH Q8 (R2) [2] using knowledge management and quality risk management, (see ICH Q9) [3].

A modern PQS is considered key to enabling the effective implementation of ICH Q8 (R2) and Q9.

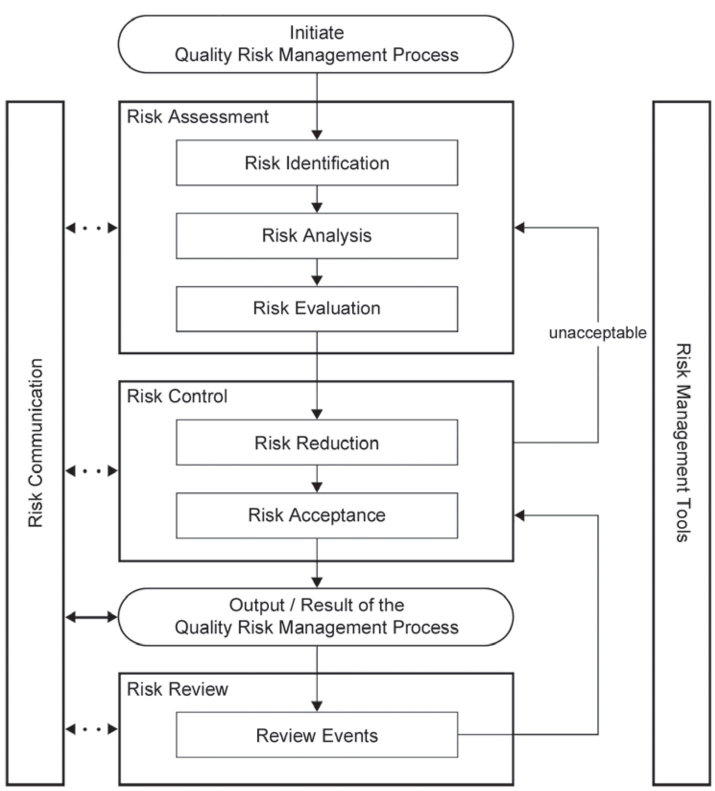
As part of the approach to support the implementation of modern quality thinking and design strategies, ICH Q8 (R2) aims to allow manufacturers to apply more effective control over the management and implementation of changes for drug products which have been developed using the principles of Q8 (R2). It also encourages the assessment of process performance and monitoring of product quality.

ICH Q8 (R2) is one in a suite of complementary ICH guides that define and encourage a modern approach to pharmaceutical quality management. It defines key concepts such as design space and control strategy, and is considered a key enabler in the acquisition, development, and transfer of knowledge for drug products.

ICH Q11 [4] provides further clarification on the principles of ICH Q8 (R2), ICH Q9, and ICH Q10 as they relate to the development and manufacture of drug substance (API).

ICH Q9 provides guidance on the principles of quality risk management and the use of quality risk management tools. These tools can enable more effective and consistent risk-based decisions across the product lifecycle (see Figure 1.1).

Figure 1.1: Overview of a Typical Quality Risk Management Process (from ICH Q9)

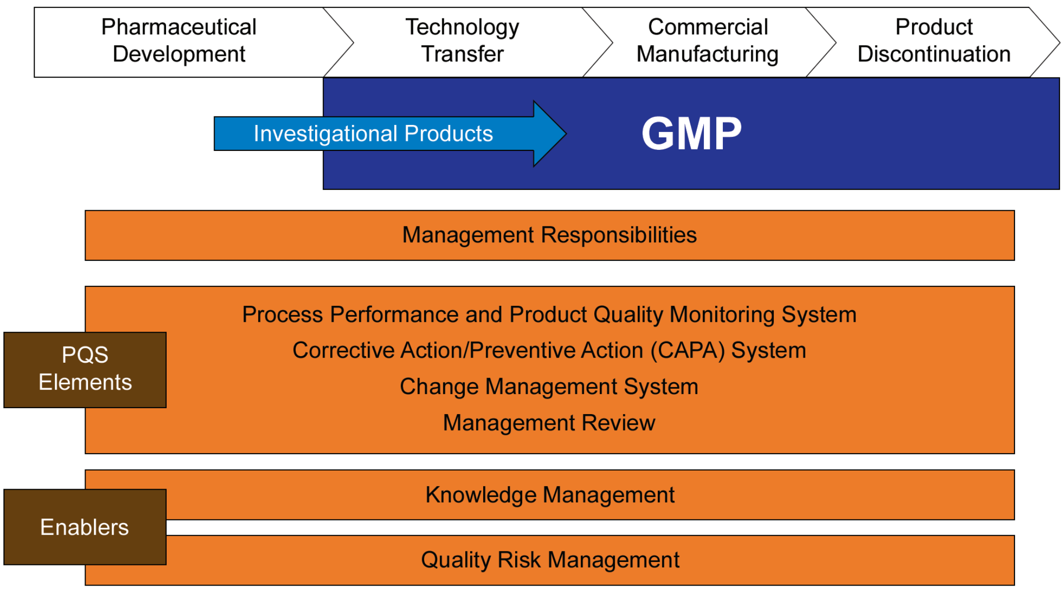


ICH Q10 describes a lifecycle approach to a PQS, from development through product discontinuation. The knowledge about a pharmaceutical product and the processes required to reliably produce that product starts with product and process development. An effective PQS uses the knowledge acquired throughout the lifecycle of the product, builds on that knowledge, and applies it to:

- Other stages of the product lifecycle
- Other product lifecycles

A change management system is an important element of a PQS as seen in Figure 1.2 – reproduced from ICH Q10, [1], which illustrates the major features of the ICH Q10 PQS model over the product lifecycle.

Figure 1.2: Diagram of the ICH Q10 Pharmaceutical Quality System (from ICH Q10)



The ICH Q10 model also describes management responsibilities and specifies the four elements of a PQS (ICH Q10 Section 3.2) as:

1. Process Performance and Product Quality Monitoring System
2. Corrective Action and Preventive Action (CAPA) System
3. Change Management System
4. Management Review of Process Performance and Product Quality

A change control system which is a GMP requirement is described in ICH Q7 [5]; however, a change management system which is discussed in ICH Q10 is considered much broader in scope.

1.2 Purpose

The purpose of this Guide as a key element of a pharmaceutical quality system is to describe potential product lifecycle approaches to the change management system element of a PQS (ICH Q10 Section 3.2). These approaches can help to achieve the opportunities described in ICH Q10.

The principles described in this Guide also can be applied to alternative PQS approaches that organizations may choose to implement.

The Guide aims to:

1. Describe different interpretations of PQSs, based on regional GMPs, ICH Q10 guidance, and ISO standards, to help meet regulatory expectations.

2. Provide clear guidance on achieving compliant change management systems.

3. Describe how to implement compliant change management systems within a PQS with example case studies.

4. Demonstrate the inter-relationship between ICH Q8 (R2), ICH Q9, and ICH Q10.

1.3 Scope

This Guide aims to provide descriptions of good practices for change management.

Good practices can apply to products and processes developed using minimal or enhanced, Quality by Design (QbD) approaches as discussed in Appendix 1 of Q8 (R2) [2]. Changes may occur as a response to a problem-solving activity or due to a corrective action resulting from an investigation, even when using the minimal QbD approach. Some form of change management system is required regardless of the type of quality system used.

Note: good practices described in this Guide also may be applied to drug substance (API) manufacture as given in the change control Section 13 of ICH Q7, Good Manufacturing Practice Guide for Active Pharmaceutical Ingredients [5].

1.4 Benefits

The benefits of this Guide as applied to products and processes developed using minimal or enhanced, QbD approaches as discussed in ICH Q8 (R2) and Q11 include:

1. Facilitating a compliant and consistent approach to the implementation of change management.

2. Provision of practical guidance on the use of ICH Q9 in the context of change management within a modern PQS.

3. Description of approaches to change management within a PQS which are consistent with GMP regulations and ISO standards.

1.5 Key Concepts

The key concepts in this Guide include:

1. The lifecycle approach, as promoted in ICH Q10; using a consistent change management model throughout.

2. The role of the change process steward and how that can be applied in national and multi-national settings.

3. ICH Q10 and ISO 9001 quality systems and how they can be applied to assist GMP compliance in a modern PQS setting.

1.6 Structure of this Guide

The Guide is structured as follows:

1. Introduction
2. Description of Alternative Models of a Quality System
3. Change Management System
4. Case Studies
5. Glossary

2 Alternative Models of a Pharmaceutical Quality System

The range of Pharmaceutical Quality Systems (PQSs) that could be implemented should be evaluated to determine which best meets the need of the respective company. These systems may range from a PQS based on CGMP requirements to a system based on ISO 9001. The PQS selected can affect prospective change management system options, as well as other features of the PQS which also may affect the change management system (see Table 2.1).

PQSs must remain in compliance with regional GMP requirements. These requirements and guidance provided in ICH Q7 provide a foundation for ICH Q10-based PQSs.

In an ISO 9001 quality system, selected business processes should be mapped and agreed. Personnel should be trained in these selected business processes. In an organization operating an ISO 9001 system, these business processes (e.g., financial management) are made transparent.

The main differences between a PQS based on ICH Q10 and a PQS based on ISO 9001 or CGMPs relate to the extent of:

- Continual improvement initiatives
- Senior management responsibilities

A change management system is a GMP requirement regardless of the PQS used, but the extent of application of a change management system will depend on the PQS.

2.1 PQS Based on CGMPs

A PQS based on CGMPs is the traditional approach operated by the pharmaceutical industry. It consists of complying with a set of regional GMPs that are applied by regulatory authorities worldwide. Although the details of GMPs may vary between regulatory authorities, the basic principle is to ensure patient safety through application of the GMPs. Organizations must comply with the GMP requirements of the countries or regions to which they are supplying product; therefore, compliance with regional GMPs sets a minimum standard for a PQS. Organizations may choose to set higher standards, e.g., in the case of ICH Q10.

2.2 PQS Based on ICH Q10

The key benefit of implementation of an ICH Q10-based PQS across a product lifecycle is that it helps to implement the Quality by Design (QbD) concepts of ICH Q8 (R2). An ICH Q10 quality system:

- Strengthens the links between pharmaceutical development and commercial manufacturing activities
- Helps to facilitate innovation and continual improvement

Organizations that implement an ICH Q10-based PQS should be able to demonstrate an improved understanding of the development and manufacturing processes that impact product quality. This insight should enable more effective evaluation of the potential impact of change and may allow organizations to make changes under their PQS, without necessarily having to file a post approval regulatory application/supplement. In addition, ICH Q10 describes the role that senior managers have in assessment and review of the PQS.

In an ICH Q10-based PQS, continual improvement is focussed on development and manufacturing processes that have a direct impact on product quality. Underlying business processes that may indirectly impact quality (e.g., the full development process, the process for financial management) may not be mapped within an ICH Q10 PQS; therefore, senior management responsibilities do not include these business processes.

2.3 PQS Based on ISO 9001

According to the ISO website, *"The ISO 9000 family of standards represents an international consensus on good quality management practices. ISO 9001:2008 is the standard that provides a set of standardized requirements for a quality management system, regardless of what the user organization does, its size, or whether it is in the private, or public sector. It is the only standard in the family against which organizations can be certified – although certification is not a compulsory requirement of the standard. The ISO 9001:2008 standard provides a tried and tested framework for taking a systematic approach to managing the organization's processes so that they consistently turn out product that satisfies customers' expectations."*

ISO 9001 is a widely adopted standard that is used as the basis for the ICH Q10 approach. In an ISO 9001 quality system, selected business processes are mapped, therefore helping to facilitate innovation and continual improvement across business processes that impact on customers' expectations, i.e., a broader scope than ICH Q10. Senior management responsibilities include review of these business processes.

Table 2.1 shows the relationship between ICH Q10 quality system features and whether these features would be achieved under the three different models of a GMP, ICH Q10, and ISO 9001-based PQS. As regulatory authorities adopt elements of ICH Q8 (R2), ICH Q9, and ICH Q10 the requirements for a PQS may move from the current GMP-based approach to an ICH Q10-based approach.

Table 2.1: Example Differences in Requirements of a PQS

Key PQS and Business Features	PQS Based on GMP	PQS Based on ICH Q10	PQS Based on ISO 9001
Outsourcing			
• Control and review of outsourced activities included in management responsibilities.	-	√	√
Change Management			
• System should be effective, ensure continual improvement, include risk management and change of product ownership, be evaluated post implementation.	-	√	√
CAPA			
• CAPA to be implemented following specified investigations and root cause to be determined.	√	√	√
Quality Risk Management	√	√	√
Enhanced, Quality by Design Approach	-	**Optional**	**Optional**
Process Performance and Product Quality Monitoring System			
• Establish control strategy to facilitate timely feedback/feed forward, measure, analyze, identify sources of variation, and drive continual improvement.	-	**ICH Q8(R2)**	**Product Realization/ ICH Q8(R2)**
Business Process Performance	-	-	√
Knowledge Management	-	√	√
Senior Management Responsibilities			
• Accountability for PQS including governance and reviews, establish Quality Policy and Quality Objectives.	-	√	√

Notes:
1. √ indicates requirements covered in the respective PQS
2. Underlined text emphasizes specific requirements, e.g., GMPs require control of outsourcing, but do not specifically mention inclusion of outsourcing control within management responsibilities.

3 Change Management System

3.1 Introduction

Change management is one of the four PQS elements described in ICH Q10. This Chapter provides guidance on how a change management system is operated within an ICH Q10-based PQS.

The change management system principles and model are described, based on ICH Q10. This model can be applied to each stage in the product lifecycle. This Chapter also describes how the enablers, knowledge management, and quality risk management can apply to each step of a change management system.

3.2 Change Management System Principles

The principles of an effective change management system apply, regardless of the type of PQS or the stage within the product lifecycle. For example, when using a minimal approach to development and a PQS which is based only on compliance with GMPs, a change management system is still required to evaluate and implement changes. In this case, changes may occur, e.g., from resolution of troubleshooting exercises or from complaint investigations. Where a pharmaceutical system is required to support products and processes developed using an enhanced, Quality by Design (QbD) approach and the principles described in ICH Q8 (R2), [2], an effective change management system is considered essential.

ICH Q10 [1] (Section 3.2.3) describes four considerations for inclusion in the change management system in relation to the stage in the lifecycle, these include:

- The potential impact of the change using quality risk management tools
- The potential impact on marketing authorizations
- Evaluation of the proposed change by subject experts
- That the change objectives were achieved

The CGMP requirements must be met regardless of the stage of the lifecycle or whether the product and processes have been developed using minimal or enhanced, QbD approaches. CGMP requirements usually include:

- A Standard Operating Procedure (SOP) which describes the approach to evaluation, approval, and implementation of the proposed change and confirmation that the change is effective and did not have negative consequences
- Clear accountabilities for all change management activities
- Prioritization of urgent and important changes (e.g., a corrective or preventive action)

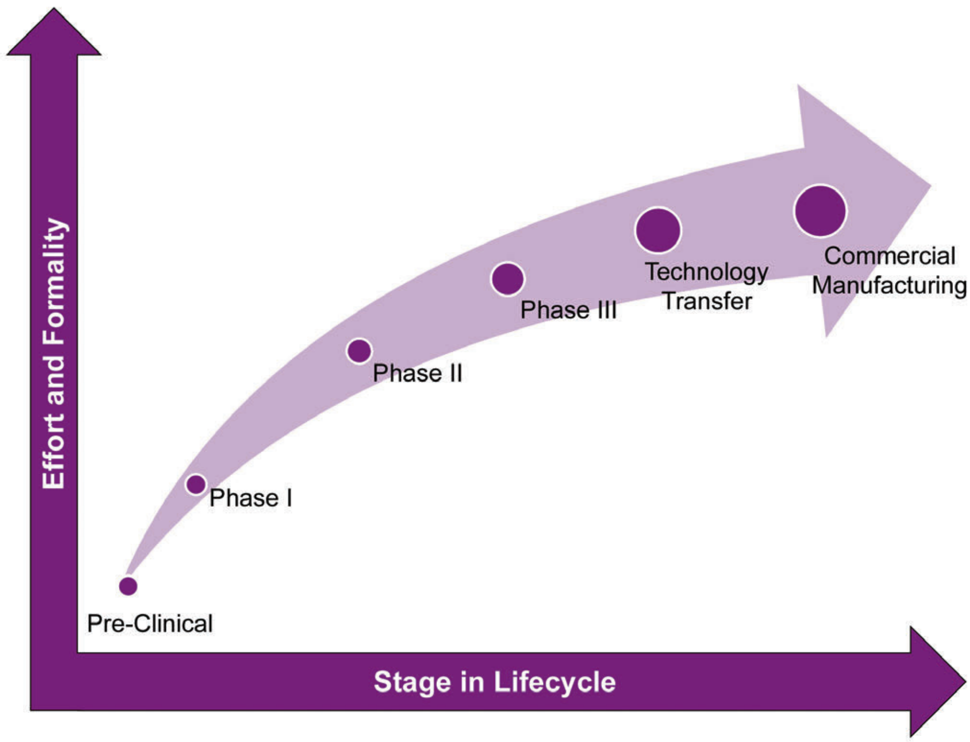
In regard to change management, the Quality Unit (QU) should:

- Ensure that a change management system is established
- Approve or reject proposed changes
- Audit the impact of changes post implementation
- Review effectiveness of the change management system and recommend improvements, where necessary

- Assure the safety, efficacy, and quality of the product that results from the change

The extent and depth of the application of the change management principles can vary according to the stage in the lifecycle where the change is proposed. For example, there may be less rigorous application of these principles during development, particularly during early stage development, than in commercial manufacture. Figure 3.1 gives a relative depiction of this. For further information on the level of formality, see Section 3.3 of this Guide.

Figure 3.1: The Level of Change Effort and Formality over the Product Lifecycle



For further information on specific considerations with respect to change management, see Section 3.4 of this Guide. (These considerations apply to organizations which have chosen to implement design space and control strategy.)

3.3 Change Management System Model

ICH Q10 defines change management as, *"A systematic approach to proposing, evaluating, approving, implementing and reviewing changes."*

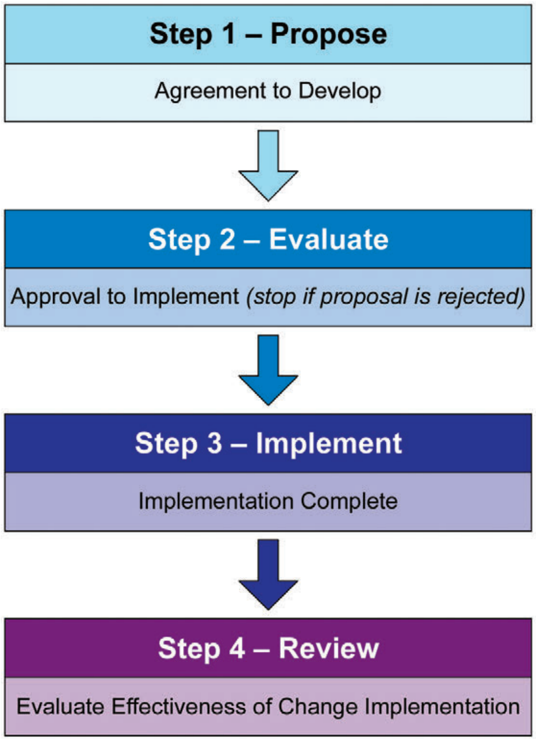
This description makes clear that the scope of change management is broader than the scope of change control, which is described in ICH Q7 [5], Section 13.

"Change control" often refers to the execution step of an individual change, whereas "change management" is a more systematic holistic approach to the review and management of a portfolio of changes and the change process, including all components of "change control." In an ICH Q10-based PQS change management applies to the entire product lifecycle. Additionally, if an organization chooses to operate an ISO 9001 system, the change management system could also include other elements not covered under ICH Q10 or GMP, e.g., changes to business processes.

The high-level model described in this Chapter is adapted from the ICH Q10 definition of change management and the four steps described in ICH Q10, see Figure 3.2. Each step ends with either a gate or a decision point which should be completed before proceeding to the next step. This Chapter also provides further information on each step within the model.

Following these steps is considered an effective way to ensure that proposed changes are evaluated and implemented effectively. It may be necessary to return to an earlier stage, e.g., if the benefits of implementation are not fully realised. The level of formality of application of these steps can vary depending on the stage in the lifecycle, see Section 3.4.

Figure 3.2: Step and Gate Model for Change Management



3.3.1 Change Management System Enablers

ICH Q10 describes two enablers for implementing ICH Q10 effectively:

1. Knowledge Management

2. Quality Risk Management

Knowledge Management

Product and process knowledge is acquired throughout the lifecycle of a product and can be stored in several formats, e.g.:

- Electronic databases

- Written reports

- Regulatory submissions

Conclusions from all development studies, including scale up and technology transfer activities supported by the data are part of product and process knowledge. Knowledge also can be gained from similar process, e.g., platform technologies and from the scientific literature.

Where applied, design space, control strategy, and regulatory commitments can be considered summaries of product and process knowledge.

Product and process knowledge can be important when proposing, evaluating, and implementing change.

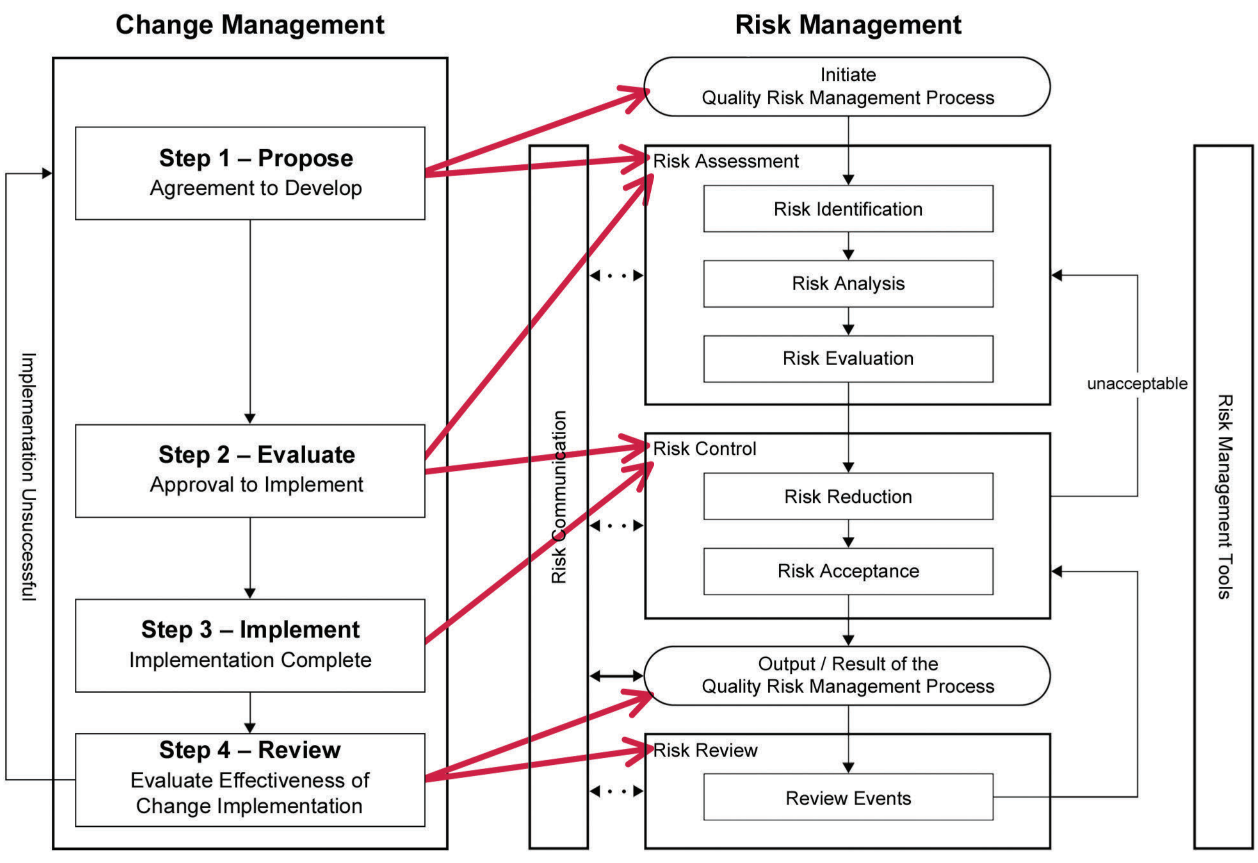
Quality Risk Management

As outlined in ICH Q9, quality risk management can be used during the steps described in Figure 3.2 to help assess:

- The proposed change
- The effectiveness of the change once implemented

Figure 3.3 illustrates how the risk and change management system could interact to facilitate an effective change management system.

Figure 3.3: Relationship between Change Management System Model and ICH Q9 Quality Risk Management Model



Note: The solid arrows point to important components of the quality risk management model that should be taken into account at the relevant stage within the change management system.

3.3.2 Executing the Change Management System Model

A Change Process Steward should be designated and should be accountable for the change management process. The role of Change Process Steward should provide a clear focal point for the:

- The management of change
- The portfolio of change proposals and plans
- The measurement of the effectiveness of change

This role of Change Process Steward may be site, regional, or international depending on the size and scope of operations. This role can be important when a change may impact several locations; in this case, the role should be supported by local change process owners.

The change management system may be part of a broader set of accountabilities relating to product quality, e.g., process performance and product quality monitoring. These accountabilities could be combined into a role called "Product Quality Steward."

The Change Process Steward also may be accountable for periodic evaluation of the effectiveness of the change management system and for the development and improvement of that system.

For manufacturers or importers located in Europe (or other regions with similar requirements), the Qualified Person (or a person with similar experience and role) may be considered as a candidate for Change Process Steward.

The four steps of the change management system model are:

- Step 1 – Change Proposal
- Step 2 – Change Evaluation
- Step 3 – Change Implementation
- Step 4 – Review: Evaluation of Implementation

Step 1 – Change Proposal

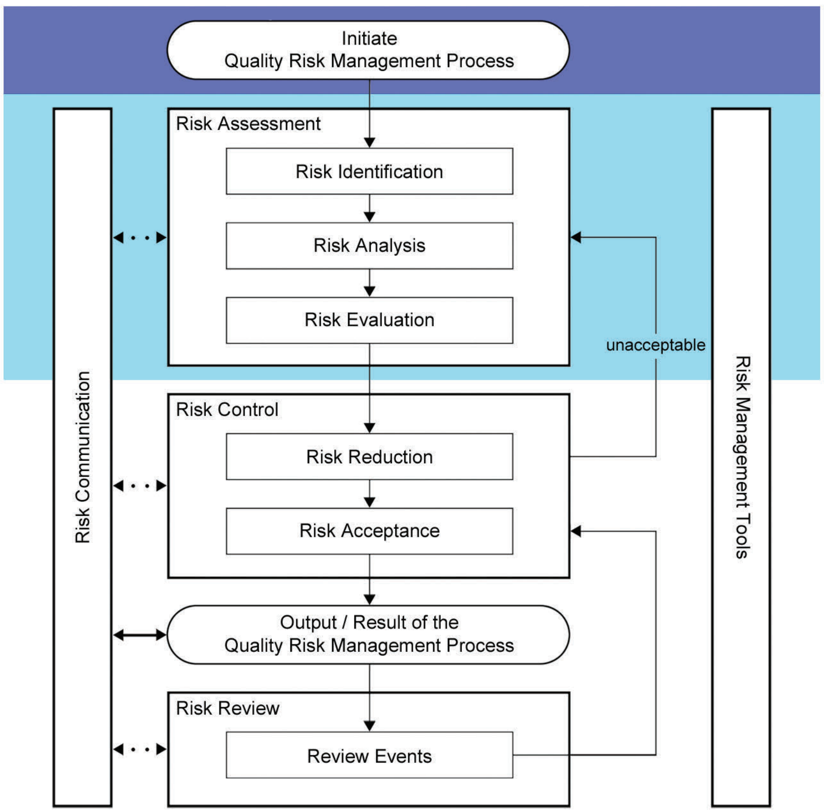
A change proposal may originate from any point within an organization and may arise from several sources, including:

- Corrective Action and Preventive Action (CAPA)
- Outcome of experiments or clinical trials in development
- Technology innovation
- Process performance and product quality monitoring
- Management reviews of process performance and product quality
- Regulatory agency interactions and inspections
- Knowledge transfer from other manufacturing locations and development
- Continual improvement activities
- Business strategy, e.g., lifecycle changes, relocation of manufacturing or R&D, outsourcing decisions
- Specific local legislative requirements

- Changes in product ownership
- Changes in internal organisational arrangements, e.g., working patterns
- Internal harmonization of regulatory expectations

Change proposals should consider the potential impact of changes on other processes, products, and locations (where the organization has multiple sites). The use of ICH Q9 quality risk assessment tools can be effective in ensuring that the potential impacts have been considered during the first step of the change management system model (see Figure 3.4).

Figure 3.4: Step 1 – Propose Change



The application of Quality Risk Management (QRM) begins with the creation of the change proposal.

A high level **risk assessment**, using informal QRM tools, can be used to:

- Identify significant risk and potential impacts
- Estimate implementation costs
- Support a high level risk/benefit analysis
- Determine whether to proceed to Step 2 – Evaluate Change

The results are summarized in the change proposal and communicated to the Change Process Steward for consideration.

A detailed **risk assessment** should only be conducted after the change has received agreement to proceed to Step 2 – Evaluate Change.

The proposer of a change should provide a high level change proposal or change request (see Section 6 for an example proforma) which can be used to determine if additional resources should be supplied to further develop the change. The change proposal should include a:

- Clear indication of the nature, risks, and benefits of the change
- Review of all relevant prior knowledge
- Estimate of resources
- Relative priority

For example, changes which will implement corrective or preventive actions from deviation or complaint investigations should be given priority over other changes.

The high level change proposal should consider the potential impact on regulatory commitments and the possible need for regulatory submissions. The impact on other processes, products, and locations (where the organization has multiple sites) also should be considered. The informal use of ICH Q9 quality risk assessment tools can be effective in ensuring that the potential impacts have been considered. A summary of the findings of the risk assessment should be included in the change proposal.

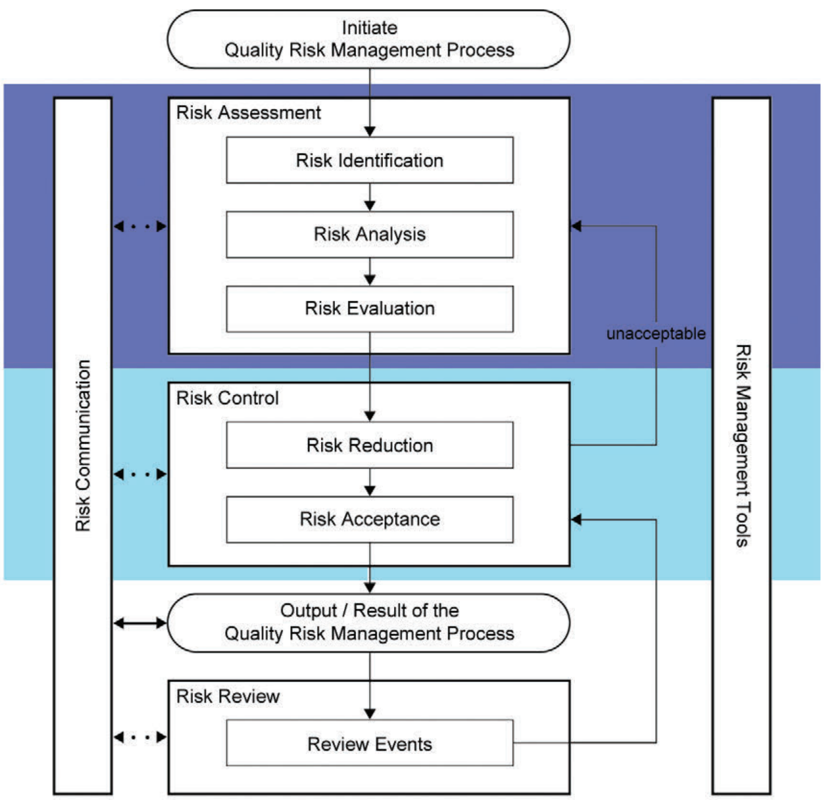
The high level proposal should be reviewed by the Change Process Steward to ensure that the change is consistent with the overall portfolio of changes and that there is sufficient budget and resources to be able to act on the change. The Change Process Steward should be aware of the available capacity of the organization to deal with the portfolio of changes and can determine the respective priority of the proposed change with respect to the portfolio. If the proposed change is considered urgent, the Change Process Steward should be able to designate the change as high priority. For more complex change proposals, it may be helpful for the Change Process Steward to consult relevant functions, departments, or individuals, e.g., members of the Review Board (see Step 2 Change Evaluation), who should support the Change Process Steward in delivering their part of the change, with the Change Process Steward acting as project manager.

Once the high level proposal is approved for further development the proposal moves to Step 2 Change Evaluation.

<u>Step 2 – Change Evaluation</u>

This stage requires a deeper assessment of the proposed change, and includes the key steps of impact evaluation and development of an implementation plan. This is considered a critical step in successful change management.

Figure 3.5: Step 2 – Evaluate Change



The most extensive use of QRM occurs during this step.

The change impact analysis is based on a detailed **risk assessment**. It should include the risks of change to:

- Quality, efficacy, and safety of product
- Other systems and equipment

Preparation of an implementation plan that should include **risk control** elements such as:

- (Re)qualification or (re)validation
- Additional testing or monitoring

Formal QRM tools such as Failure Mode Effects Analysis (FMEA) may be used for complex or critical changes. The risks and controls are communicated in the more detailed change proposal.

Following agreement to proceed, the change proposal moves to Step 3 – Change Implementation.

To facilitate this step, a more detailed change proposal (or change control) should be prepared, to enable a decision to be made regarding acceptance or rejection of the proposal. This change proposal should include:

- Description and scope
- Justification
- Evaluation of the impact of the change
- Implementation plan
- Change implementation success criteria

The proposal should then be assessed to determine whether to proceed to implementation.

Description and Scope

- Provides a high level description of the change
- Considers whether the change is temporary or permanent. If the change is intended to be temporary. the end date for the change should be specified.
- Considers the impact on regulatory submissions, equipment, buildings, products, and analytical methods, etc.
- Considers the potential impact on third parties

Justification

- Describes the reason the change is being requested and the rationale for using the resources
- Describes any alternative approaches that have been considered, if applicable
- The justification can form part of the basis for evaluation of the success of the implementation of the change

Evaluation of the Impact of the Change

A thorough impact evaluation should be performed as it is considered critical to successful change management, particularly for complex changes. The formal risk assessment steps in ICH Q9 and the ICH Q9 tools may be used to evaluate the potential impact of the changes and identify the risks associated with the implementation.

The types of issues which should be considered during an evaluation of the risks and impact on product quality, safety, and efficacy of a proposed change include:

- GMP compliance, including validation state
- Design space (including any impact on critical process parameters and critical quality attributes)
- Control strategy (including any impact on critical process parameters, critical quality attributes, product specifications, and product release)
- Analytical methodology and/or quality control standards
- Additional process knowledge on this, similar products, or similar processes
- Process performance and product quality monitoring

- Product stability and whether additional or extended stability trials are required
- Environment , health and safety, and business operations
- Registration status and the need to update or seek approval for changes in registration files
- Amendments to SOPs, batch documentation, master files, etc.
- Computer validation status, including software and hardware
- Maintenance and metrology protocols
- Downstream customers including third parties (e.g., packing sites)
- Changes in the scope and scale of a facility, including downsizing and upsizing and potential impact on the quality system
- Other locations, including third parties which are impacted by the change
- Any prior knowledge that has been accrued through the implementation of other changes whether they were successful or unsuccessful

Implementation Plan

After the risks and impacts have been determined, the risk control steps described in ICH Q9 can be used to develop the implementation plan. The application of risk control is considered an effective way of ensuring that all relevant considerations have been taken into account. The risk controls can vary depending on the stage of the product lifecycle, e.g., in the commercial manufacturing stage of the lifecycle, risk controls may include qualification or validation activities.

The content and amount of detail in the implementation plan depends on the stage of the lifecycle and the criticality of the proposed change. In cases where multiple regulatory approvals are required, the implementation plan should have an approach to maintaining continued regulatory compliance.

Change Implementation Success Criteria

Criteria for measuring the success of the implemented change should be established as part of the implementation plan. Examples of success criteria include:

- Confirmation the change accomplished the goals which were included in the justification
- Avoidance of harms (negative consequences) identified in the risk assessment, e.g., increased impurity levels in an API
- Avoidance of unforeseen negative consequences, e.g., the change was the root cause of a deviation
- Confirmation of regulatory approval, where required
- Completion within cost and time requirements

Obtain Approval to Implement

There may be several key stakeholders that should be consulted before a decision is made to proceed with the change.

A multi-disciplinary Change Management Review Board/Team ("Review Board"), which has the accountability for assessing individual changes and how these changes fit with the change portfolio can help to ensure that appropriate stakeholders have been consulted. This Review Board should include appropriate representation to make an effective assessment of the proposed change and the implementation plan and success criteria. The membership and size of the Review Board can vary from organization-to-organization and should reflect the size of the operations and stage in the product lifecycle. The Review Board may hold virtual meetings or make assessments independently using on-line tools; this can be particularly effective for organizations that exist in multiple locations.

A typical Review Board should be comprised of technical Subject Matter Experts (SMEs) who have decision-making rights for the functions that they represent, and are able to evaluate change proposals. Depending on the size of operations, the stage in the product lifecycle, and the potential impact of the proposed changes, a Review Board may include appropriate representation from:

- Production
- Quality Assurance
- Quality Control
- Development
- Engineering
- Technical Support
- Regulatory
- *Ad-hoc* members (e.g., statistical support, finance), as appropriate

The Change Process Steward should chair the Review Board.

Where several sites may be impacted, approval should be obtained from the appropriate Change Process Owner at each site.

In Europe (or other regions with similar requirements), the Review Board also should include the QP or designate. In the US, there should be input from the Quality Unit and in Japan from the Good Quality Practice Group.

The Review Board is normally responsible for deciding whether the proposal is accepted, rejected, or held (e.g., pending further information). This decision should be recorded in writing, and signed off by the Quality Unit.

The Review Board should communicate the outcome of the change evaluation to the proposer with a clear indication of reasons for further work or rejection when either of these decisions is reached.

On completion of these steps and approval to implement, the change moves to Step 3 Change Implementation. This signifies the transition from change management to change control, i.e., execution of the change, and commits resources to implement the change.

Step 3 – Change Implementation

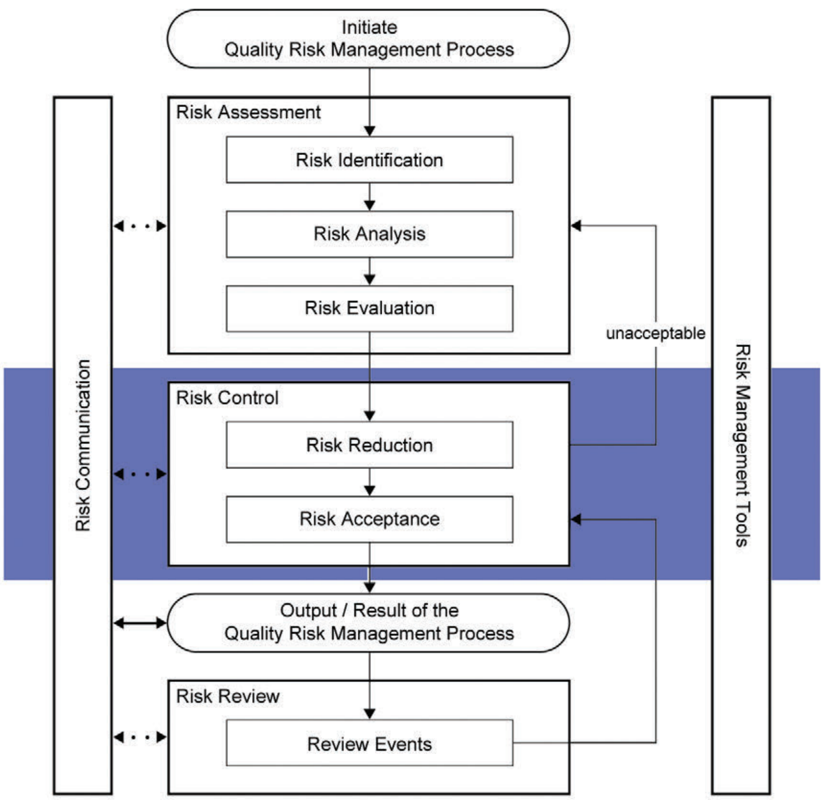
From this step, standard project management techniques should be applied. The key concepts of Project Management are included in the ISPE Good Practice Guide: Project Management [6].

Once the Review Board accepts the proposal, a project manager should be appointed. The Project Manager should be responsible for executing the implementation plan and for evaluation of the effectiveness of the implementation of the change.

The Project Manager should update the implementation plan, as appropriate to the stage of the product lifecycle. For example, a detailed project plan should be prepared for a change that impacts commercial manufacturing, whereas for a change in early development it is appropriate to document the nature of the change and when it will take place. The nature of the change should determine whether a cross-functional project team is required or whether the change can be delivered by an individual or function.

A well executed risk assessment in the change proposal (see Step 1 Change Proposal) and coupled with a good implementation plan from Step 2 Change Evaluation, should help to avoid unanticipated negative consequences when implementing the change. The risk controls which were planned using the ICH Q9 tools should be implemented as part of change implementation. These controls may either reduce risks or monitor for unacceptable consequences. Additionally, risk management documents such as Failure Mode Effects Analysis (FMEA) may need to be revised to document the implementation of these additional risk control activities. It also may be appropriate to communicate short term risks and risk reduction precautions to impacted areas during implementation.

Figure 3.6: Step 3 – Implement Change



The **output/result** from the QRM in the previous steps are now implemented.

Risk controls may include:

- Revising operating parameters
- Installing new equipment
- Monitoring for potential consequences of accepted risks
- Creating or updating SOPs and training
- Updating regulatory submissions

It also may be appropriate to:

- Create or revise QRM knowledge documents such as an FMEA
- Communicate short term risks and risk reduction precautions to employees in impacted areas

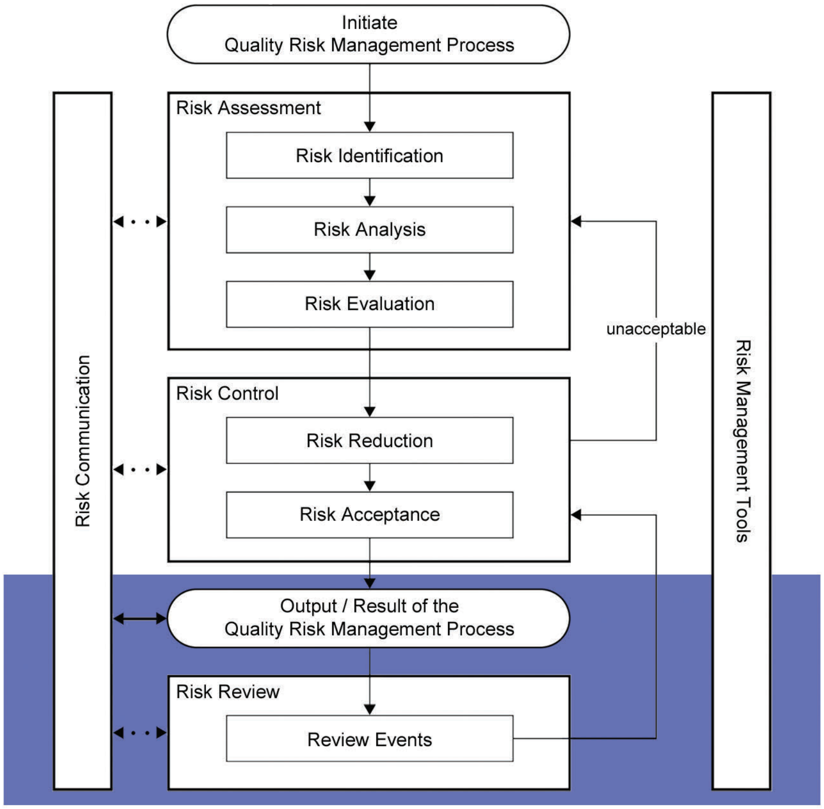
Appropriate training should be included in the project plan.

If the success criteria or feasibility of the project plan causes the initial proposal to be re-considered, the plan should be referred to the Review Board and evaluated as described in Step 2 Change Evaluation. Significant changes to the approved change scope or implementation plan should be formally documented and reapproved.

Step 4 – Review: Evaluation of Implementation

The project manager should confirm that all steps in the implementation plan have been completed, and evaluate the effectiveness of implementation of the change against the success criteria. This evaluation should include the output of the quality risk management process.

Figure 3.7: Step 4 – Review the Completed Implementation of the Change



The final portion of QRM begins when the implemented change is reviewed. This will confirm the **output/result of the QRM process** as follows:

- Verify execution of the entire implementation plan, including plans for updating QRM tools, risk controls, and QRM communication.
- **Review events** associated with the change implementation. Also consider the success criteria and if any negative events were reported

The other elements of the PQS provide an ongoing QRM **risk review** after closure of the change. Examples include the results of product review, inspections, audits, and deviation management.

On completion of the evaluation, the project manager should report the outcome to the Review Board. This should allow retention of knowledge and provide knowledge for dissemination by appropriate representatives to their respective functions. It should include the transfer of successful and unsuccessful change implementation for future reference and development of prospective change proposals, i.e., should be incorporated in the body of knowledge for subsequent related change proposals.

In the case of unsuccessful change implementation, the Project Manager may recommend that the change be referred to one of the earlier steps in the change management system model (See Figure 3.2 Step and Gate Model for Change Management).

Considerations to ensure an effective review of the implementation of the change include:

- Sufficient data points, as described in the implementation plan, should be gathered to the timelines described, before an assessment of the change is made.

- The success criteria referred to in Step 2 Change Evaluation should be achieved. If not, reasons why they have not been achieved should be assessed along with the mitigation steps to address the reasons why, including reverting to the previous operating state where appropriate. This may require the proposal of a subsequent change or amendment of the implementation plan to ensure success, i.e., referral back to an earlier stage of the Step and Gate Model for Change Management (See Figure 3.2).

- Data and knowledge gathered from implementation of the change should be shared with the development function and other locations, as appropriate, to ensure that learning can be applied in products under development or to similar products manufactured at the same or other locations.

- Where the implementation of a change is unsuccessful, the reasons for failure should be clearly understood and acted upon.

After closure of the change, other elements of the PQS, particularly the process performance and product quality monitoring system and the management review of the change management quality system, should be used to monitor the impact of the change. If these systems detect change management issues, the CAPA system can be employed to determine and take action to correct the root causes.

Documentation and Knowledge Management

- The change management process that is operated should be documented in organization Standard Operating Procedures (SOPs) and these SOPs should be available for external inspection or internal audit
- There should be appropriate supporting data to justify the change and the impact on product quality.
- Documentation should be retained, according to the GxP requirements of the markets supplied or to the document retention requirements of the organization, whichever is the longer. This documentation should be available for external inspection or internal audit.
- It is recommended that all of the documents relating to each change are retained in a distinct file or there is a clear audit trail to demonstrate the inter-connection between all of the steps in the change management process.
- Where regulatory approval of a change has been required, internal documentation should be updated to reflect changes to approved regulatory commitments.
- Changes should be documented in product quality reviews/annual product reviews as appropriate for the stage of the lifecycle.
- Knowledge generated from evaluation and implementation of any change should be transferred to the development or manufacturing functions as appropriate to enable greater product and process understanding that can be applied to other processes or products. This should be included in the accountabilities of the Change Process Steward.

3.4 Change Management in the Product Lifecycle

Although the basic change management system should be the same for all stages of the product lifecycle, differences in application of the change management system model may be relevant, depending on the stage. This Section describes the specific stages in the product lifecycle and how the change management processes described in this GPG can be applied. Case studies are included in Appendix 2.

3.4.1 Change Management in Pharmaceutical Development

For the purposes of this Guide, the pharmaceutical development part of the product lifecycle includes:

- Drug substance (API) development
- Formulation development (including container/closure system)
- Manufacture and supply of investigational products
- Delivery system development (where relevant)
- Analytical method development

- Manufacturing process development and scale-up

The manufacture of investigational medicinal products is governed by regional GMP requirements.

A key goal of pharmaceutical development is to design a product and associated processes, e.g., manufacturing, that consistently deliver product of the requisite quality, safety, and efficacy to patients in clinical trials, and potentially commercial supply. For generic manufacturers, the extent of development is less significant than for the development of new chemical entities. Much pre-clinical to Phase 3 work is conducted by the innovator. Regardless of the stage of development, there is a need to ensure that a suitably robust drug product which meets the appropriate standards of quality, safety, and efficacy is supplied to patients taking part in clinical trials. Depending on the phase of development, further pre-clinical and/or clinical development may be required.

Change is an inherent part of the development process and should be documented; the formality of application of the change management process should be consistent with the stage of pharmaceutical development.

Pre-Clinical

Development of new products includes preclinical safety evaluation and clinical trials in Phases 1, 2, and 3. Many potential candidates for new products do not survive the preclinical safety evaluation. As such, change in pre-clinical safety is governed by the basic scientific principle of good record keeping, and this is outside the scope of this Guide. As products progress through the development part of the lifecycle, fewer and fewer will make the transition into later development phases. Pre-clinical studies, while outside the scope of this Guide, provide valuable prior knowledge as an input to subsequent stages of the product lifecycle.

Phase 1

Products are administered to a small number of human volunteers for the first time during Phase 1 and the principles of GxP and ICH Q10 apply. Apart from the important pre-clinical knowledge, there is relatively limited knowledge about the compound that is administered. For Phase 1 trials, it is commonplace to use a simple formulation, e.g., suspension or liquid.

At this stage, the change management process should still be relatively informal, with a review of any potential patient safety implications, GMP compliance, and the impact on any regulatory filings for the clinical trials, e.g., IND and CTA being the key considerations. Typically, a limited number of changes are made <u>during</u> Phases 1 and 2; however, based on the knowledge acquired during Phase 1 trials, changes are frequently made prior to the start of Phase 2. The steps described in Figure 3.2 should be followed, but there is no need to go through an extensive process. The proposed change should be evaluated, recorded, and signed off by the Quality Unit before implementation. Changes may be incorporated in the Development plan.

Phase 2

Following evaluation of Phase 1 results, a decision should be made whether to proceed with the next stage of development. Phase 2 trials involve an increased number of patients. As more knowledge is acquired during Phase 2, additional changes may be proposed, e.g., for subsequent clinical trial applications. Some organizations may choose to use prototype versions of the formulation and analytical methods that will be used in Phase 3 and ultimately commercial manufacture, although these prototypes are more commonly used for Phase 3 trials. At this stage, the change management process should still be relatively informal, with a review of any potential patient safety implications, GMP compliance, and the impact on any regulatory filings for the clinical trials, e.g., IND, CTA. The steps described in Figure 3.2 should be followed. Since there are now implications for a greater number of patients, and evolution to the final formulation is usually underway, greater consideration should be given to the implications of any change that is proposed. Although the process should remain relatively informal, there should be greater input from stakeholders, and greater emphasis should be placed on stage 2. The proposed change should be evaluated, recorded, and then signed off by the Quality Unit. Changes may be incorporated in the Development plan.

Phase 3

Phase 3 trials involve a significant number of patients. Following the evaluation of Phase 2 results, a decision is made whether to proceed with the Phase 3 stage of development. Based on the knowledge acquired during Phase 1 and 2 trials, changes are usually made prior to the start of Phase 3. As more knowledge is acquired during Phase 3, additional changes may be proposed, e.g., for subsequent clinical trials or commercial manufacture.

Typically, the final formulation and analytical methods are used for Phase 3 trials, although late stage prototypes also may be used. In some organizations, the facilities that may be used for commercial manufacture are used to provide the clinical trial supplies for Phase 3. Alternatively, Phase 3 clinical trial manufacture may be conducted in development facilities and then transferred to the commercial manufacturing facilities as part of the technology transfer stage. In both cases, consideration should be given to the potential impact on any clinical trial or commercial products that are co-located in the same facility.

The expectation of regulatory authorities is that the standards of GMP applied are higher than the earlier phases and approximate to the standards in commercial manufacture. The level of formality of application of the change management system described in Figure 3.2 and Section 3 should now be much greater than for the earlier phases of development. Where clinical trial supplies for Phase 3 and commercial products are manufactured in the same facility, consideration should be given to having a single change management process with sufficient flexibility to address the specific risks associated with each type of activity.

The particular considerations for change during Phase 3 of development include:

Step 1 – Change Proposal

- It is routine to use the facilities that are used for commercial manufacture or distribution. In this case, a key consideration of any change proposal is the potential impact on any of these operations, and hence the extent of stakeholder interaction, e.g., through the Review Board.

- The success criteria should include the feasibility of implementing the change in later stages of the lifecycle.

- The impact on any regulatory filings for the clinical trials, e.g., IND, CTA should be considered.

- A development team or technology transfer may take on the role of Review Board, with appropriate stakeholder interactions where there is no representation on the team.

Step 2 – Change Evaluation

- The impact on the bioavailability in comparison to other earlier formulations, e.g., through changes to synthesis of the API

- The extent to which subsequent modifications to analytical methods result in methods that are at least comparable to methods used in earlier phases, e.g., in detecting impurities that are generic to the formulations developed

- The impact of modifying package components and potential impact on the safety and purity of the investigational product

Step 3 – Change Implementation

- The change implementation plan should be incorporated in, or be an integral component of the overall development plan.

- An evaluation of implementation should be conducted as per Section 3.

- The change implementation evaluation should include the feasibility of implementing successful change in the technology transfer arrangements for commercial manufacture.

3.4.2 Change Management in Technology Transfer

Technology transfer may occur during or after Phase 3 development, and subsequent technology transfers may take place to additional commercial manufacturing locations, including to third party contractors through site-to-site transfers.

The purpose of technology transfer is to transfer all of the relevant knowledge acquired during the development stage of the product lifecycle to the sites that will undertake the commercial manufacture, packing, analysis, and distribution of medicines to achieve product realization. This knowledge forms the basis for the manufacturing process, control strategy, process validation approach, and ongoing continual improvement. For site-to-site transfer of commercial manufacture, the knowledge transferred is enhanced by experience of running the commercial manufacturing and associated processes. Technology transfer is a unique type of change in that it usually involves multiple coordinated changes. The change management system should manage and document any adjustments, e.g., to the manufacturing process, that are made during technology transfer activities.

At this point, the GMP standards expected by the regulatory authorities are the same as for commercial manufacture. The change management system described in Figure 3.2 and Section 3.3 are applicable in full to technology transfer. There are particular considerations for the change management system during technology transfer, i.e.:

Step 1 – Change Proposal

- Development of the technology transfer strategy, which should incorporate all of the proposed changes to successfully initiate commercial manufacturing.

- The impact on any emerging , filed, or approved regulatory filings, e.g., MAA, NDA, JNDA

- The success criteria, including any changes to plant configuration and validation status, knowledge transfer back to development for subsequent transfers

- Review of the control strategy

- The level of detail should reflect the complexity of the technology transfer

Step 2 – Change Evaluation

- The impact on any site specific information including site master files and other relevant regulatory information

- The impact on master documentation e.g., batch records, metrology , test methodology

- The impact on site services, e.g., water and air

Step 3 – Change Implementation

- The change implementation plan should be incorporated in or be an integral component of the technology transfer strategy.

Step 4 – Review: Evaluation of Implementation

- The change implementation evaluation should assess the validation status of the facility post technology transfer, including facility and product validation.

- The closure of the change proposals
- Completion of relevant documentation, e.g., batch records and creation of certificates of analysis
- Publication of risk management reports
- Summary technology transfer report that captures the status of all of the proposed changes
- Amendments to emerging, filed, or approved regulatory filings, e.g., MAA, NDA, JNDA
- Effectiveness of the control strategy
- "Snag list" of any changes that need further attention

3.4.3 Change Management in Commercial Manufacturing

This section provides practical guidance regarding the management of change in a commercial manufacturing environment, and includes all changes that have the potential to impact the quality of commercially available products. It does not include any subsequent steps to notify or gain the approval of the regulatory authorities where the change has the potential to impact on registration files.

In a commercial manufacturing environment, the extent of the controls that are exercised to manage change are much more rigorous than in other phases of the product lifecycle described in ICH Q10. At this point in the product lifecycle, the commercial manufacturing processes, test methods, control strategy and batch release processes have been approved by regulatory authorities and products are being supplied to patients. Any proposed change needs to be assessed for potential impact on safety, quality, and efficacy, and against the GMPs and regulatory files for the territories that are intended to be supplied after the change has been made. These requirements are more onerous for commercial manufacture. Full application of the change management system depicted in Figure 3.2 and described in Section 3.3 is recommended.

The particular considerations for commercial manufacturing include:

Step 1 – Change Proposal

- The impact on other commercial and development products that are manufactured, packed, or tested in the same facility
- The impact on any approved regulatory filings, e.g., MAA, NDA, JNDA for the products which are the subject of the change, and for other products that are located in the same facility
- Any differences in the status or extent of regulatory filings, e.g., different countries may have different requirements for the manufacturing description or files may not yet have been approved
- The impact on other facilities that manufacture, pack, or test products that are the subject of the change
- The impact on upstream activities, e.g., component and raw material supplies
- The impact on downstream activities, e.g., packing, distribution

Step 2 – Change Evaluation

- Detailed evaluation of the items described in the change proposal is normally required in order to create a detailed implementation plan.

- The impact on the design space, if applied, including any impact on critical process parameters, critical quality attributes, and underlying assumptions
- The impact on the control strategy, including any impact on critical process parameters, critical quality attributes, the associated acceptance criteria, product specifications, and product release processes
- The need for new and revision of existing risk assessments and existing risk controls or monitoring processes as well as determining if additional risk controls are required
- The impact to equipment qualification, process validation, and continued process verification
- The impact on any site specific information including site master files and other relevant regulatory information
- The impact on master documentation, e.g., batch records, metrology, test methodology
- The impact on site services, e.g., water and air
- The impact on product quality, including stability
- The risk of being unable to meet market demand in full or in part during change implementation
- The impact on SOPs and personnel qualification and training

Step 3 – Change Implementation

- The change implementation plan is typically a stand-alone plan that may include multiple changes.
- To ensure the change will be implemented as approved, the plan should provide for documented verification of the completion of key steps or milestones and for any required training of affected individuals,
- Implementation should include the transfer of knowledge to development or to other locations manufacturing, packing, or testing the same products.

Step 4 – Review: Evaluation of Implementation

- Assess or confirm the validation status of the facility post implementation of the change including facility and product validation
- Include enough data points to statistically determine if the change met the predetermined success criteria
- Evaluate if the change had unanticipated negative consequences, such as a deleterious impact on product quality
- Continue monitoring the long term impact of implemented changes using the process performance and product quality monitoring system

3.4.4 Change Management in Product Discontinuation Phase

Product discontinuation is considered to be the removal of a product from commercial supply, i.e., the complete cessation of all manufacturing, release, and supply activities. If the product is to be transferred or sold to another manufacturing unit, including third parties, see Section 3.4.2 on change during technology transfer. Product discontinuation signifies the end of the Q10 lifecycle.

The change model outlined in Section 3.3 also should be followed for product discontinuation, and the steps in the model should be managed in the same way.

Particular consideration should be given to:

- Consultation with regulatory authorities prior to cessation to ensure that there are appropriate medicines available to substitute for the product removed

- Stability programs still running at the time of discontinuation. They should continue until they are concluded, and the same principles of assessment and reporting apply.

- Communication to the markets impacted regarding the timing of the discontinuation should be made in good time to allow alternative medicines to be sourced.

- Retention of documentation in accordance with the GMP requirements of the market supplied

- Retention of product and technology know-how should supply re-commence in the future

- Communication to users of the product or their proxies, e.g., specialists or general practitioners

- Controlled withdrawal of product and/or diversion to alternative products

- Any legal or product liability issues that may arise as a result of the product discontinuation

These considerations should apply even in cases when product is recalled or removed from the market at the same time as product discontinuation or if supply is being run down to a point in the future.

3.5 Change Management of Design Space and Control Strategy

The design space and control strategy summarize key knowledge and controls to provide assurance of quality. During change management, they should be reviewed from two perspectives:

- They are useful sources of knowledge to evaluate the impact of proposed changes

- They should be kept current

The change evaluation should include determination if design space and/or control strategy changes will also be required.

Table 3.1, which describes the evolution of design space and the control strategy throughout the lifecycle, is intended to help consider the potential implications for those organizations that have chosen to implement a design space. For further information on control strategy and design space topics, see the ISPE Guide: PQLI® from Concept to Continual Improvement, Part 1 – Product Realization using QbD: Concepts and Principles" [7].

Table 3.1: Control Strategy through ICH Q10 Lifecycle

	Product Lifecycle			
	Pharmaceutical Development	**Technology Transfer**	**Commercial Manufacturing**	**Product Discontinuation**
Define Parameters to Control	Dimensional Analysis\Scale-Up/ Scale Down Design Qualification (DQ)	Risk Management Process Modeling Operational Qualification (OQ)	Process Qualification (PQ)	Not Applicable
Establish Control Methods, Design Space and Monitoring Mechanisms	Measurement Strategy Experimentation Trial and Error DoE	Critical Process Parameters Process Modeling	Multivariate SPC Simulation Neural Networks Alarm Systems	Adverse Events Customer Complaints State IV Clinical Stability
Implement Control Strategy	Process Engineering User Requirements Specification (URS)	Process Modeling Functional Requirements Specification (FRS)	Process Capability Process Control Standard Procedures (SOPs) Control Plan	Post Market Surveillance for Marketed Product Annual Product Review
Continual Improvement	New Formulations Reverse Engineering Re-Formulations QbD/PAT	Characterization Monte-Carlo Analysis	Process Variability Reduction Process Waste Process Review Process Optimization	Critical Input into New or Improved Product Formulations

Any proposed change should be assessed for the potential impact on safety, quality, and efficacy and against the GMPs and regulatory files for the countries supplied, as described in Section 3.3 of this Guide. These principles apply to any proposed change to the design space and/or control strategy where they are operated.

In most regions, working within the design space is not considered a change that requires regulatory approval; however, the proposed change must still go through the change management system as required by GMP to ensure effective evaluation and subsequent implementation.

If the design space or elements of it are included in regulatory filings, it may be necessary to gain regulatory approval prior to implementation of the change. Therefore, a key consideration during evaluation is the commitments that have been made through regulatory filings.

Where the design space or elements of it are not included in regulatory filings, movement within the design space is considered to be under the direct control of the organization managing the change and there is normally no need to consult regulatory authorities regarding the change. In this instance, the proposed change should still go through the change management system, but there is no need to go through a regulatory approval step.

In either case, maintenance of the validated state of the design space is a key consideration in ensuring effective implementation of the change as described in Section 3.3.

Movement outside of the design space, e.g., expansion of the design space, is considered to be a change and would normally initiate a regulatory post approval change process. The knowledge gathered from any changes to the design space should be transferred to development and other locations who manufacture the same product.

The impact on control strategy also should be evaluated. If it is planned to amend the control strategy, an assessment should be made regarding any potential impact on product quality, safety, and efficacy, and on the control strategy summary that is approved by a regulatory agency. This is a key consideration if the change proceeds to implementation and ensures that the necessary regulatory approvals are in place.

In some instances, additional elements of the control strategy are adopted by manufacturers that do not appear in the control strategy summary that is filed with regulatory authorities. In these cases, the changes should still go through the change management steps with a particular emphasis on ensuring the effective implementation of the new operating parameters and associated documentation to maintain the validated state of the control strategy.

Appendix 1

High Level Change Proposal or Change Request

4 Appendix 1 – High Level Change Proposal or Change Request

Change Control Number:		**Date:**
Background Information		
Stage of Lifecycle Impacted by the Change:	Region/Country Impacted by the Change:	Originator of the Change:
Change Control Summary		
Current Description (pre-change):		
Desired Outcome (post-change):		
Justification/Reason for Change:		
Priority (High/Medium/Low):		
Gap Analysis (difference between current and desired state):		
High Level Risk Assessment (GMP/Safety, Environment/Technical/Registration/Design Space/Control Strategy):		
Benefits of Change:		
Estimate of Resources Required (specify Functions):		
Signed:		Approved:

Appendix 2

ICH Q10 Product Lifecycle Case Studies

5 Appendix 2 – ICH Q10 Product Lifecycle Case Studies

5.1 Pharmaceutical Development

1. Change Proposal

The Change Proposal was for a change in route design for API manufacturing for a development project between Phase 2b and Phase 3 clinical study. The rationale for the change was two-fold:

1. The avoidance of an explosive intermediate during the synthesis of the API
2. A substantial reduction in the cost of goods

2. Change Evaluation

The risk assessment indicated that there were potential risks relating to human volunteer safety and the regulatory filings that had been made to regulatory authorities:

- Introduction of any new unidentified impurities during synthesis of the new route
- Validity of the analytical method to determine purity and detect impurities not observed in the old synthetic route
- Introduction of potential genotoxic impurities, e.g., by using new starting materials or production of by-products or degradants
- Not being able to demonstrate equivalency between the API from the superseded and new synthetic routes, and ultimately bio-equivalence of the formulation for the Phase 3 studies
- Generation of the correct documentation is needed to support this change for Phase 3
- Generation of documentation to support the change and evaluation of the impact of the change for the New Drug Application (NDA)

To mitigate these risks, appropriate controls were incorporated into the implementation plan:

- Characterization of the API from the new route including determination of the impurity profile and identification of each impurity together with comprehensive analysis of the crystal morphology in order to address concerns over bio-equivalence
- Review and re-validation of the analytical method
- Assessment of the potential genotoxic impurities using the new route, including all starting materials and intermediates
- A one-month toxicology study to compare with the previous batches of the API produced by the old route to qualify the new impurity profile to support Phase 3 Investigational New Drug (IND) and Clinical Trial Applications (CTA)

The implementation plan was reviewed by the development team and incorporated into the overall development plan before being approved by the Quality Unit.

3. Change Implementation

The change was implemented according to the implementation plan described above. The specific risk controls were tracked throughout the implementation to ensure that they were successful. The change was implemented prior to the start of Phase 3 and reviewed before the Phase 3 clinical trials commenced. Analytical methods were updated and re-validated to ensure that previously unseen impurities could be detected. The SOPs for the plant manufacturing were updated to enable the manufacture of the clinical drug to the new synthetic route.

4. Review: Evaluation of Implementation

After implementation, the change was reviewed by the development team. The development team concluded:

- The API manufactured by the new route was well characterized.
- Physico-chemical equivalence of the API from the new synthetic route was demonstrated.
- New impurities did not present a toxicological risk to patients.
- The analytical method was successfully re-validated and detected the new impurities.

The development team approved the use of the API in the Phase 3 clinical trials. The data generated during implementation of the change was included in the package to support Phase 3 Investigational New Drug (IND) and Clinical Trial Applications (CTA); these were approved by the regulatory agencies of the countries where the trials were planned to be conducted.

5.2 Technology Transfer

1. Change Proposal

The proposal was to transfer the technology for a new product from the development site to the commercial manufacturing site.

2. Change Evaluation

The risk assessment for the transfer did not indicate any specific areas of concern, particularly as the unit operations concerned were simple and well-characterized. Therefore, there was no reason to believe that this unit operation would present any risk to product quality and patient safety.

The implementation plan for the change, which included pre-defined acceptance criteria for the blending unit operation and dosage uniformity, was approved by the technology transfer team and approved by the Quality Unit.

3. Change Implementation

During validation, it became apparent during material transfer that the blend uniformity step did not meet the acceptance criteria. The variability presented a potential risk to the patient, because if it had not been detected, the patient may have received variable doses that may not have controlled their medical condition.

Validation was stopped and an additional risk assessment was conducted. This detailed risk assessment on the blending unit operation evaluated all of the possible sources of material segregation, including blending, material transfer, and downstream activities. The outcome of this risk assessment identified areas of potential optimization to improve blend uniformity to meet the acceptance criteria.

Following the risk assessment, a proposal for change was referred back to the technology transfer team, i.e., back to Step 1 Change Proposal. The potential improvements that were identified during the risk assessment were included in the implementation plan for the change during Step 2 Change Evaluation and were approved to proceed to Step 3 Change Implementation. The implementation plan also included the original acceptance criteria for blending and dosage uniformity.

During implementation, the validation of the blending unit operation proceeded successfully and was implemented into the final commercial manufacturing process.

4. Review: Evaluation of Implementation

After completion of the validation, the effectiveness of the implementation the change was evaluated and the key learning points were captured:

- The risk assessment gave an effective forward view of potential risks that were likely be encountered and was the basis of the approval to implement change. The general learning obtained was that evaluation during implementation of a change is a critically important feature of change management.

- Following up with a more in-depth risk assessment using the knowledge acquired regarding the outcome of the blending unit operation enabled more effective change management to be used for the re-validation step in the second round of changes.

- Knowledge acquisition from a unit operation is not static: in this case, the technology transfer of a new product resulted in additional, unexpected knowledge that can be applied to future changes. This knowledge was transferred to both the development and commercial manufacturing functions for them to include as a consideration in their change management processes.

- The flexibility of the change management process enabled a further change to be evaluated and in this case, to be implemented successfully.

5.3 Commercial Manufacturing

1. Change Proposal

The proposal was to select additional sources of an excipient that controlled release of drug in a sustained release formulation. Additional sources of the excipient were required to meet an anticipated increase in product demand, and three alternatives needed to be evaluated. The alternative sources of the excipient met pharmacopeial standards

2. Change Evaluation

The risk assessment and impact analysis for the proposal addressed three specific areas. The first was that a design space had been described for the commercial product and the different alternative excipient supplies needed to be evaluated against the design space and control strategy to ensure that the required release profile could be achieved consistently.

The second area was to evaluate the impact of any changes to the design space on regulatory filings. The assessment confirmed that the markets supplied with product from this plant had different expectations regarding design space and changes within it; therefore, regulatory filings would be required in some markets.

Finally, the sustained release product was manufactured in multiple locations and the knowledge needed to be transferred to alternative sites and back to the development function for future developments.

The implementation plan for the change therefore included the following risk controls:

- Determination of the impact on the design space critical process parameters, notably the coating steps; the critical quality attributes, notably the functional performance of the excipient; impact on the product release profile and stability of the product

- Preparation of a regulatory action plan to update the regulatory files for the markets that were supplied

- Knowledge transfer plan to ensure that the knowledge was transferred to the other manufacturing locations and the development function

The implementation plan was reviewed by the Change Management Review Board, who required an amendment to the plan to ensure that the knowledge transfer plan initiated a change proposal at the other manufacturing locations to ensure any differences in plant configuration or practices were accommodated at those locations. They also requested increased process and product monitoring for the first ten batches using excipient from each new source. The implementation plan was subsequently signed off by the Change Management Review Board and approved by the Quality Unit.

3. Change Implementation

During implementation, the assessment of the impact on the design space confirmed that only one alternative supply of the excipient could be adopted without making amendments to the design space. In this case, to achieve the required product release profile, changes to the coating parameters were required within the design space. The plant operating procedures were updated accordingly.

To adopt the remaining two alternative supplies, the design space and control strategy would need to be developed further and the coating process would need to be revalidated. It was agreed that to pursue these alternatives, separate change proposals should be generated and routed through the change management system.

The regulatory action plan was initiated with a specific requirement to use the existing supplies of excipient for the markets which required regulatory approval before the alternative excipient could be used. The quality control and plant procedures were updated accordingly.

The knowledge was transferred to the other manufacturing locations and to development in line with the knowledge transfer plan. The knowledge was subsequently applied to a product in development.

4. Review: Evaluation of Implementation

The post-implementation review confirmed the following learning points:

- The design space was a powerful tool in assessing the impact of the change and had enabled commercial manufacture to proceed with confidence in an alternative excipient. Stability test data and increased process performance monitoring confirmed that this change, which was within the existing design space and control strategy, had no negative impact on product quality.

- The other two alternative excipients showed promise and the use of the change management system was an excellent approach in ensuring that the additional work to extend the design space was undertaken.

- The knowledge was transferred effectively to development and used in a subsequent development. This confirmed that the knowledge acquired in commercial manufacturing could be beneficial in development and ultimately result in more robust commercial manufacturing processes

The knowledge transfer to other locations was only partially successful. The differences in plant configurations meant that additional work at these locations was required to implement the alternative excipient. This highlighted the need to have a corporate strategy on plant equipment and configuration.

5.4 Product Discontinuation

1. Change Proposal

The proposal for change was to discontinue a product that was supplied to multiple markets. The reason for product discontinuation was that complex manufacturing issues were identified which prevented the product from being manufactured to a suitable standard of quality and robustness. A highly skilled and experienced team investigated the issues, but failed to positively identify or resolve the root cause, resulting in the need to discontinue the manufacture of the product.

2. Change Evaluation

The risk assessment identified two significant risks that needed to be controlled. The first was that patients were using product and the potential impact of recalling product at the same time as discontinuing product supply. The second was the longer term impact on patients due to product discontinuation and whether alternative products were available to meet the needs of the patient.

The evaluation of the two key risks was reviewed as part of Step 2 Change Evaluation. It was agreed that the first risk should be accepted because the product had met the appropriate specifications and GMP standards and there was no evidence to suggest an impact on patients, e.g., reviewing complaint profiles. As the investigation into the manufacturing issues had confirmed that no root cause could be established, and that there was no likelihood of resolution the product should be discontinued. An approach to the reduction in the second risk was agreed through the substitution of an alternative product. Key steps in reducing this risk were to discuss possible alternatives with the Health Authorities of the markets that were impacted, and the feasibility of alternative suppliers being able to meet market demand.

The evaluation confirmed that in the majority of markets, there were alternative supplies available to meet demand with no risk to the patient. However, in a minority of markets there was insufficient supply to meet patient demands. In these markets, further supplies of product that was to be discontinued were made in agreement with the regulators to allow alternate suppliers time to increase supply to replace the discontinued product.

Having established an implementation plan and clear success criteria relating to the availability of medicines to patients, the Change Management Review Board, consisting of representatives from R&D, Clinical Patient Safety, Product Management, Legal, Regulatory, Quality, Supply Chain, and Communications approved the recommendation to proceed to implementation.

3. Change Implementation

The change was implemented in less than six months where alternative product was available. In the remaining markets, product was supplied in agreement with the Health Authorities over a nine month period to meet patient demand and to allow the suppliers of alternative product to increase supply to meet the demand.

The implementation plan was tracked to ensure that the change met patient needs. Additional short-term supplies of the discontinued product were supplied in agreement with the Health Authorities where necessary. Regular communications were provided to patients, health professionals, and to the Health Authorities of the markets concerned.

4. Review: Evaluation of Implementation

Implementation was achieved successfully and the product that was discontinued was replaced with alternative products in all of the markets concerned. The review identified the following learning points:

- For a global product, it is very difficult to assess the exact market penetration in each country to identify which countries would have most concern with the discontinuation. For any future discontinuations, contingency supply approaches should be incorporated in the evaluation and implement steps.

- The project team should be prepared to respond quickly to country specific issues raised by the discontinuation. Further supply may be required in some markets to allow competitors to step in or increase volume of supply.

- Product discontinuation is a complex multidisciplinary process and a standard operating procedure would have been beneficial for the project team to follow.

- Identification of the decision makers within the supply organization in relation to the decision to discontinue was a critical success factor in successful implementation. Identification of the stakeholders is an important feature in Step 2 Change Evaluation

Appendix 3
References

6 Appendix 3 – References

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Appendix 4

Glossary

7 Appendix 4 – Glossary

7.1 Acronyms

API	Active Pharmaceutical Ingredient
CAPA	Corrective Action and Preventive Action
CGMP	Current Good Manufacturing Practice
CTA	Clinical Trial Application
FMEA	Failure Mode and Effects Analysis
GMP	Good Manufacturing Practice
GxP	Good x Regulations
ICH	International Conference on Harmonisation
IND	Investigational New Drug
ISO	International Organization for Standardization
JNDA	Japan New Drug Application
MAA	Marketing Authorization Application
NDA	New Drug Application
PQS	Pharmaceutical Quality System
QP	Qualified Person
QU	Quality Unit
R&D	Research and Development
SME	Subject Matter Expert
SOP	Standard Operating Procedure
WHO	World Health Organization

7.2 Definitions

Change

Introduction of new, or a modification to existing, validated equipment, manufacturing and business processes, analytical methods, documentation specifications, raw and starting materials, components, automated/ computerized systems, facilities, storage and transport conditions, design space and control strategy.

Changes may be classified according to their impact and to the countries/markets affected (as a local, regional or global change.)

Change Control (EU GMP)

A formal system by which qualified representatives of appropriate disciplines review proposed or actual changes that might affect the validated status of facilities, systems, equipment or processes. The intent is to determine the need for action that would ensure and document that the system is maintained in a validated state. (EU GMP Annex 15 4 Glossary – 2007 book, but no longer in the online version.)

Change Control (WHO)

A formal system by which qualified representatives of appropriate disciplines review proposed or actual changes that might affect a validated status. The intent is to determine the need for action that would ensure that the system is maintained in a validated state. (WHO Quality Assurance of Pharmaceuticals, Volume 2 glossary)

Change Management (ICH Q10)

A systematic approach to proposing, evaluating, approving, implementing, and reviewing changes.

Change Management Review Board/Team (Review Board)

A multi-disciplinary team which has the accountability for assessing individual change proposals and how these changes fit with the change portfolio. Membership depends on the stage in the lifecycle and could include representation from Production, Quality Assurance, Quality Control, Development, Engineering, Technical support, Regulatory, and as appropriate ad-hoc members (e.g., statistical support, finance).

Change Model

This is the systematic and rigorous application of the model described in this guide. The model is derived directly from the key steps referred to in ICH Q10 in the definition of change management.

Change Process Steward

A person who is accountable for the change management process. This role provides a clear focal point for the management of change, including the portfolio of change proposals and plans, coordination with change process stewards/change process owner at other sites or regions, and measurement of the effectiveness of change.

Control Strategy (ICH Q10)

A planned set of controls, derived from current product and process understanding that ensures process performance and product quality. The controls can include parameters and attributes related to drug substance (API) and drug product materials and components, facility and equipment operating conditions, in-process controls, finished product specifications, and the associated methods and frequency of monitoring and control.

Design Space (ICH Q8 (R2))

The multidimensional combination and interaction of input variables (e.g., material attributes) and process parameters that have been demonstrated to provide assurance of quality. Working within the design space is not considered as a change. Movement out of the design space is considered to be a change and would normally initiate a regulatory post approval change process. Design space is proposed by the applicant and is subject to regulatory assessment and approval.

General Practitioner

A medical doctor.

Knowledge Management (ICH Q10)

Systematic approach to acquiring, analysing, storing, and disseminating information related to products, manufacturing processes and components.

Mapped

A business process covered by the relevant PQS.

Pharmaceutical Quality System (PQS) (ICH Q10)

Management system to direct and control a pharmaceutical organization with regard to quality

Quality Risk Management (ICH Q9)

A systematic process for the assessment, control, communication and review of risks to the quality of the drug (medicinal) product across the product lifecycle

Stakeholder

An individual or organization, who has an interest in the results of a change, or whose interests may be affected as a result of implementation of a change.

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